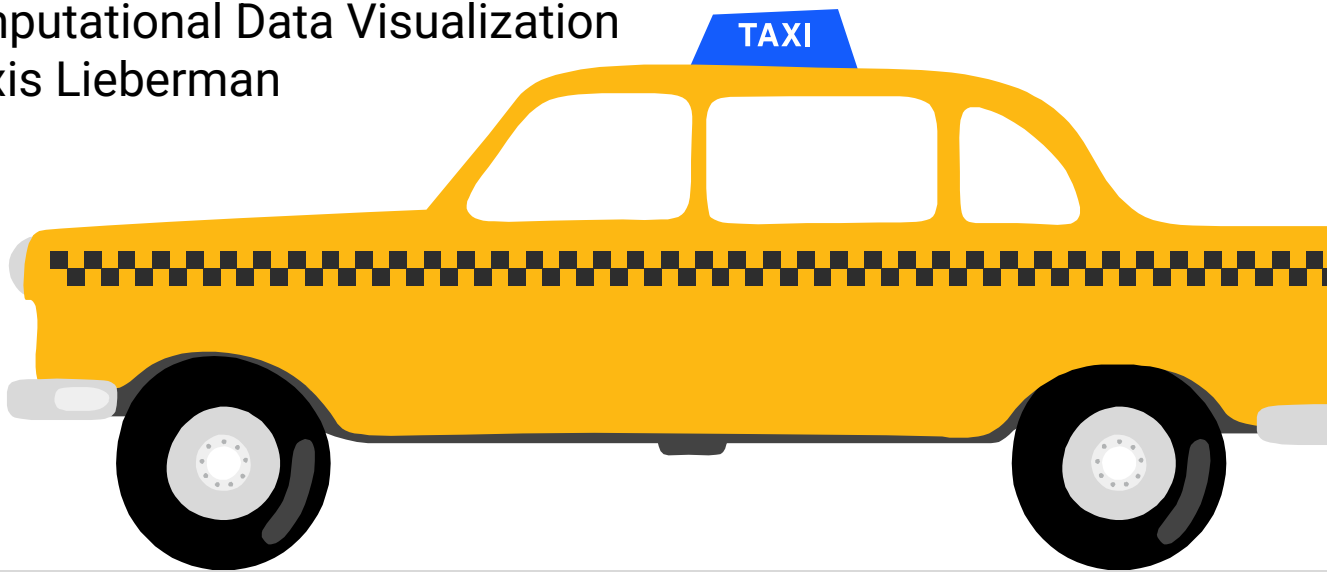
NYC Yellow Taxi Trip Dataset

CS2300 – Intro to Computational Data Visualization

Matthew Lind and Alexis Lieberman



When are taxi rides most in demand and how does the time and location of the trips affect the total amount charged?



Workflow

- Exploration of Data
- Cleaning
- Add Columns/combine data
    - Taxi Zone Lookup
    - NYC GeoJSON dataset
- Visualize
- Find Conclusions

Dataset Overview

- Information from all yellow taxi trips in New York City.
    - Variables such a pickup and dropoff time, pickup and dropoff location ID, passenger count payment type, fare amount, trip distance, etc.

- Subsetted to June and July 2025, which housed a total of 8,221,923 trips between the two months and 20 variables

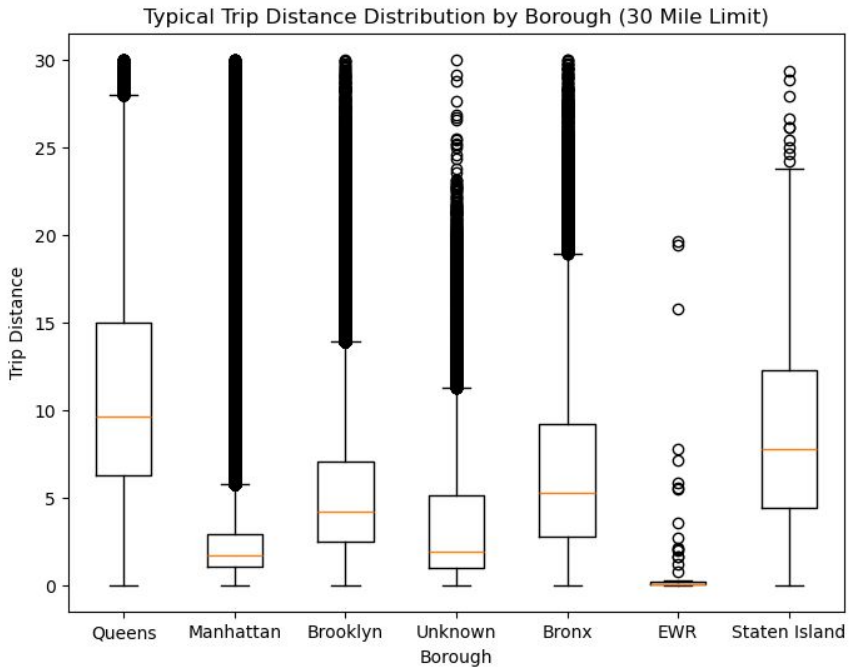
- For finding borough from Location IDs, we used the taxi zones lookup CSV from the TLC records

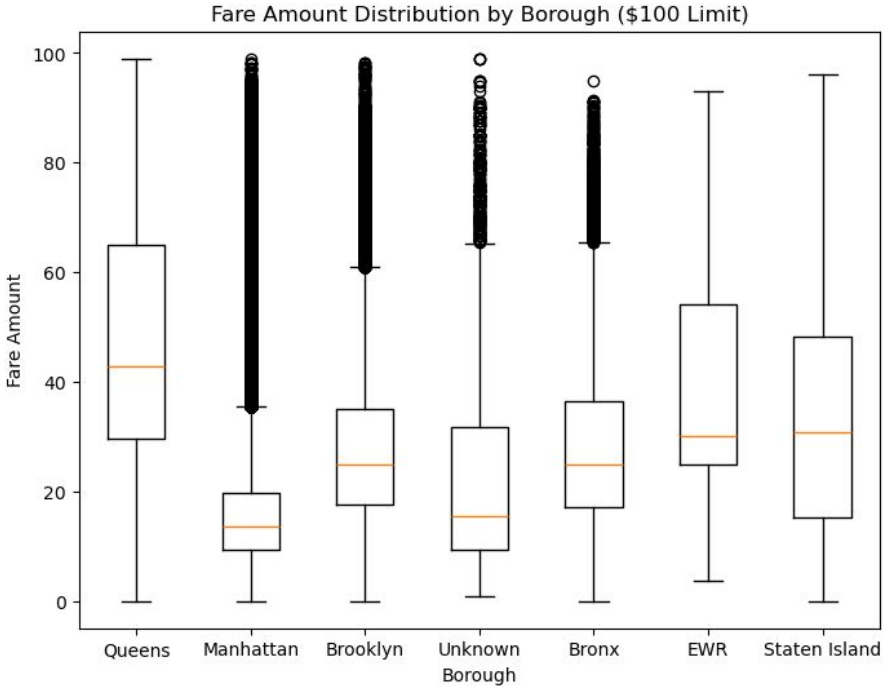
Additional data:

- For the Geospatial component, the NYC Taxi Zones GeoJSON file from NYC OpenData.
    - Columns for Shape Geometry, Shape Columns, Shape Area, Zone Name, LocationID, and Borough for 263 NYC taxi zones.

Exploratory Graphs

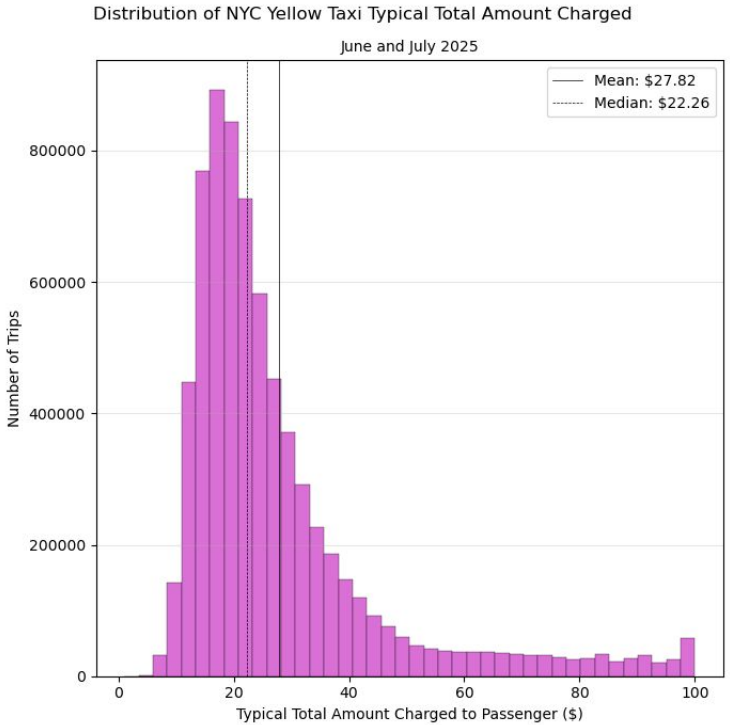
Comparing Distributions

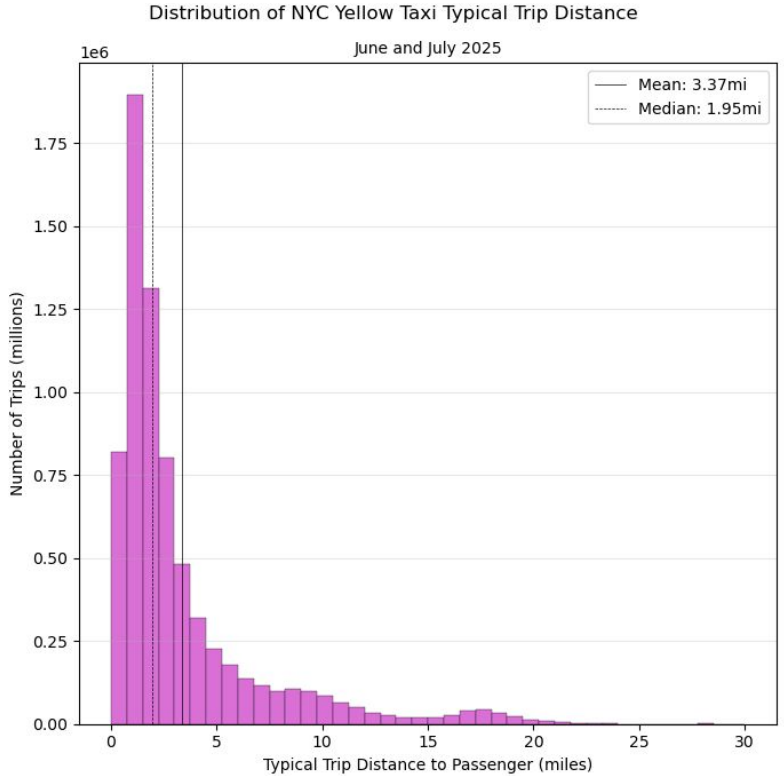




Final Visualizations

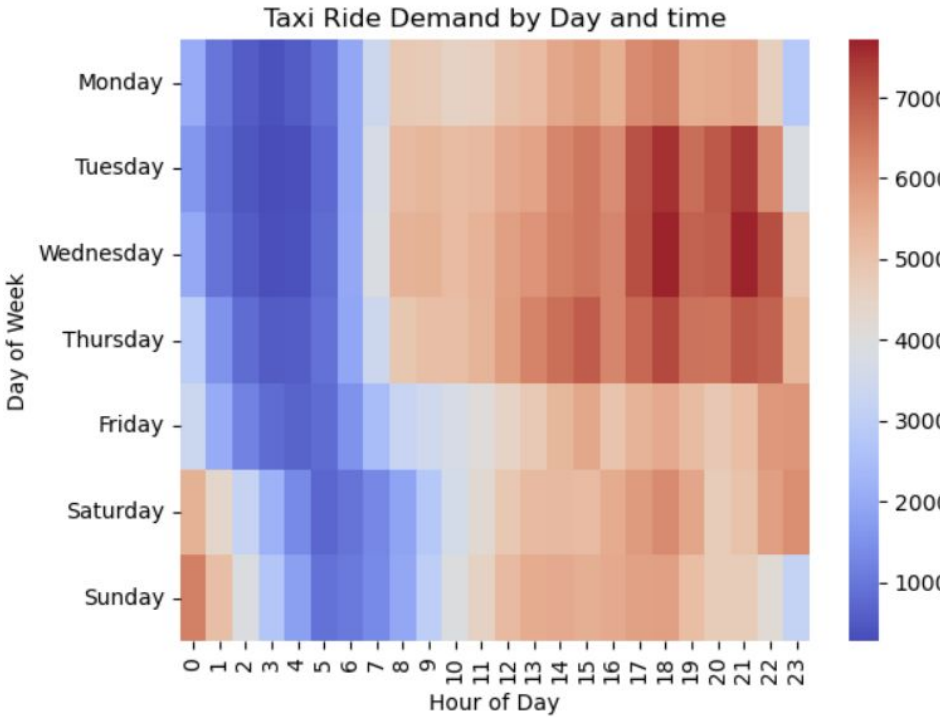
Fare and Distance





Final Visualizations

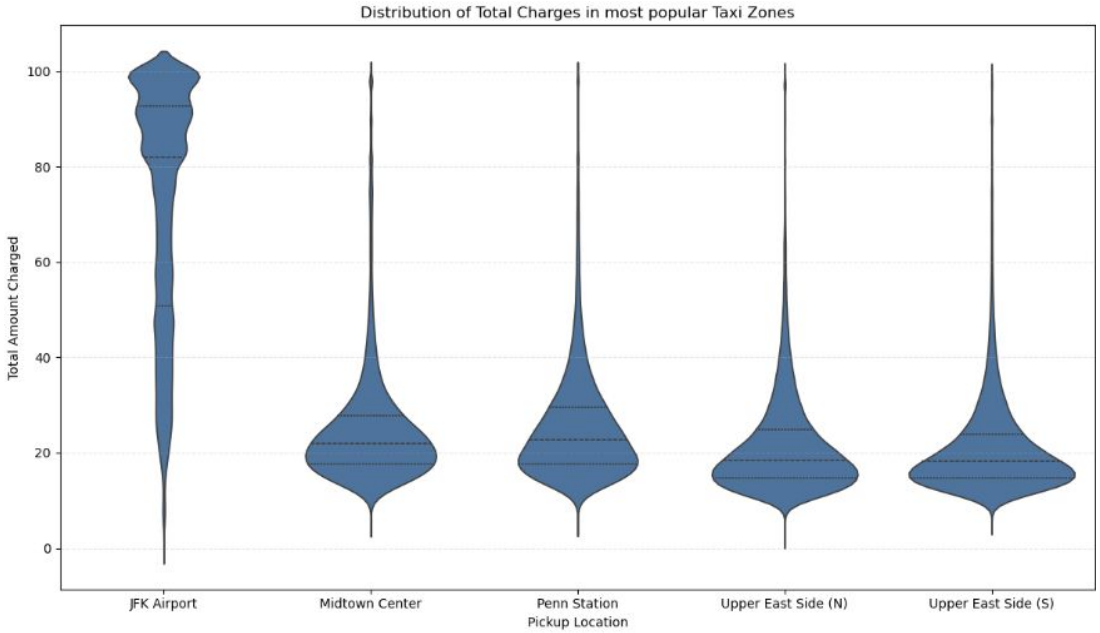
Time of Day and Week versus Demand



- You can see a “dead period” during weekday mornings
- You can see the rush hour traffic after work
- Weekdays and weekends both follow their own different matterns

Final Visualizations

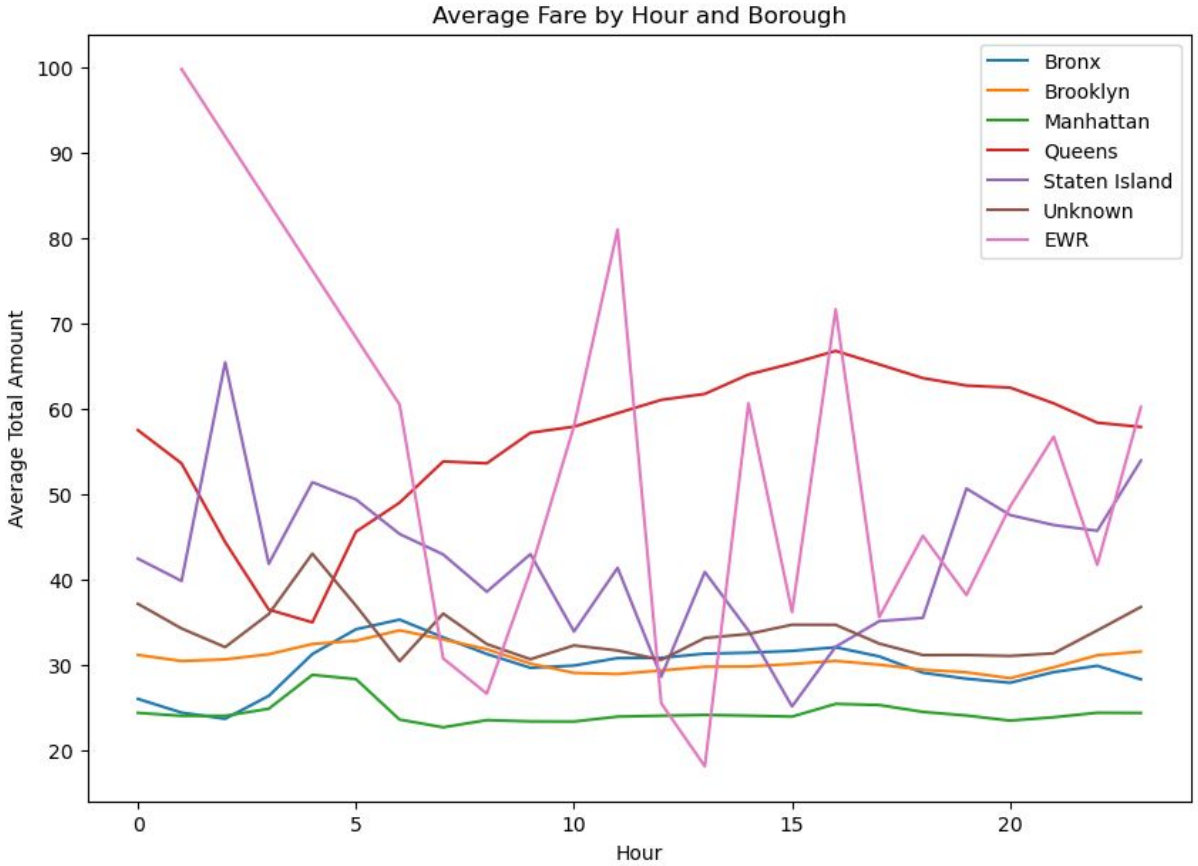
Distribution of Total Charge



- The airport has a noticeably higher total charges
- Other areas are relatively all the same

INTERACTIVE

Fare versus Hour of Day



- High variation in EWR, likely due to variation in distance (furthest borough).
- Manhattan consistently has lowest fare amount.
- Queens follows a typical rush hour pattern.

Conclusion

- Taxi fares are consistent across all of New York City except for airport zones and Queens
    - Queens is the largest borough and has 2 airports

- Taxi demand follows the typical rush hour pattern and has more ‘dead periods’ in the morning, where finding a taxi should be quite easy, but trying to get one coming home from work may not be.

- Manhattan has the largest number of trips by far, but the lowest average distance and charge.

- Fares are most dependent on distance, yet both time and location, specifically airport trips or afternoon trips also appear to have significant effects.

